

## Real Analysis II - MAT 2305

No of Credits - 03

Prerequisite - MAT 1205

Contents ;

Chapter 01 - Infinite Series

Chapter 02 - Power Series

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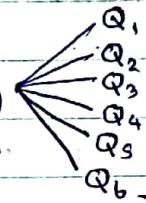
Chapter 05 - Metric Spaces.

Grading System ;

Tutorials = 20%

Mid Exam = 20%

End Exam = 60% (3 hours)



## Chapter 01; INFINITE SERIES

$\mathbb{N} = \{1, 2, 3, \dots\}$  - The set of all natural number.

$\{a_n\} \leftarrow$  Sequence

$(S_n) \leftarrow$  A new Sequence from  $\{a_n\}$

$$S_1 = a_1$$

$$S_2 = a_1 + a_2$$

$$S_3 = a_1 + a_2 + a_3$$

$\vdots$

$\vdots$

$$S_n = a_1 + a_2 + a_3 + \dots + a_n$$

$$\text{Let } a_n = \frac{1}{n}, \quad n \in \mathbb{N}$$

$$S_1 = a_1 = \frac{1}{1} = 1$$

$$S_2 = a_1 + a_2 = \frac{1}{1} + \frac{1}{2} = \frac{3}{2}$$

$$S_3 = a_1 + a_2 + a_3 = \frac{1}{1} + \frac{1}{2} + \frac{1}{3} = \frac{11}{6}$$

Let  $(a_n)$  be a sequence of real number. we associate in other sequence  $(S_n)$  with  $(a_n)$  such that,

$$S_n = a_1 + a_2 + a_3 + \dots + a_n$$

$$= \sum_{k=1}^n a_k \quad \forall n \in \mathbb{N}$$

The sequence  $S_n$  is called an infinite series (or just a series) and it is denoted by the expression  $\sum_{n=1}^{\infty} a_n$ .

$$(S_n) = a_1, a_1 + a_2, a_1 + a_2 + a_3, \dots, \sum_{k=1}^n a_k, \dots \leftarrow \text{Series.}$$

$S_n$  is called the  $n^{\text{th}}$  partial sum of the infinite series.

The infinite series  $\sum_{n=1}^{\infty} a_n$  is said to be convergent or be divergent.

If the sequence  $(S_n)$  converges to  $S$ , then  $S$  is called the sum of the infinite series. and we write.

$$\sum_{n=1}^{\infty} a_n = S \leftarrow \text{A unique value.}$$

$$(S_n) = (a_1, \sum_{k=1}^2 a_k, \sum_{k=1}^3 a_k, \sum_{k=1}^4 a_k, \dots, \sum_{k=1}^n a_k, \sum_{k=1}^{n+1} a_k, \dots)$$

i.e.

$$\sum_{n=1}^{\infty} a_n = S = \lim_{n \rightarrow \infty} S_n = \lim_{n \rightarrow \infty} \sum_{k=1}^n a_k$$

Ex 01:- Prove that  $\sum_{n=1}^{\infty} \frac{1}{n(n+1)} = 1$

Let  $a_n = \frac{1}{n(n+1)} \quad \forall n \in \mathbb{N}$

$$S_1 = a_1 = \frac{1}{1 \cdot 2}$$

$$S_2 = a_1 + a_2 = \frac{1}{1 \cdot 2} + \frac{1}{2 \cdot 3}$$

$$S_3 = a_1 + a_2 + a_3 = \frac{1}{1 \cdot 2} + \frac{1}{2 \cdot 3} + \frac{1}{3 \cdot 4}$$

$$\text{Then } S_n = a_1 + a_2 + a_3 + \dots + a_n = \sum_{k=1}^n a_k$$

$$= \sum_{k=1}^n \frac{1}{k(k+1)}$$

$$a_n = \frac{1}{n} - \frac{1}{n+1} \quad (\text{having decomposed into partial fraction})$$

$$\begin{array}{l}
 a_1 = \frac{1}{1} - \frac{1}{2} \\
 a_2 = \frac{1}{2} - \frac{1}{3} \\
 a_3 = \frac{1}{3} - \frac{1}{4} \\
 \vdots \\
 a_{n-1} = \frac{1}{n-1} - \frac{1}{n} \\
 a_n = \frac{1}{n} - \frac{1}{n+1}
 \end{array}$$

Adding all the  $n^{\text{th}}$  equation gives,

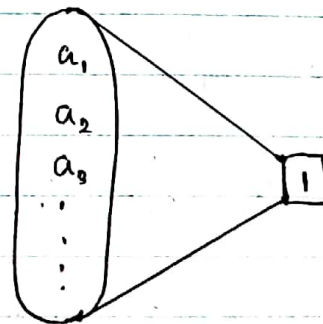
$$\sum_{k=1}^n a_k = 1 - \frac{1}{n+1} = S_n$$

$$\lim_{n \rightarrow \infty} S_n = 1$$

$\therefore$  The Series  $\sum_{n=1}^{\infty} a_n$  is Convergent and has the Sum equal to 1

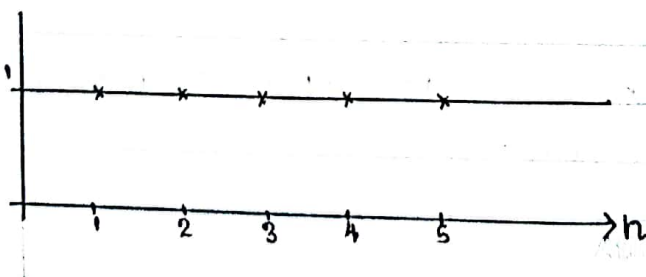
i.e.,

$$\sum_{n=1}^{\infty} \frac{1}{n(n+1)} = 1$$



Ex 02 :- Consider  $\sum_{n=1}^{\infty} 1$

Let  $a_n = 1 \quad \forall n \in \mathbb{N}$





Then,

$$\begin{aligned} S_n &= a_1 + a_2 + a_3 + \dots + a_n \\ &= 1 + 1 + 1 + \dots + 1 \\ &= n \end{aligned}$$

$$\lim_{n \rightarrow \infty} S_n \Rightarrow \lim_{n \rightarrow \infty} n = \infty$$

$\therefore (S_n)$  is divergent and so  $\sum_{n=1}^{\infty} 1$  is divergent.

Ex 03 :- Consider  $\sum_{n=1}^{\infty} 0$

Let  $a_n = 0 \quad \forall n \in \mathbb{N}$

$$\begin{aligned} \text{Then } S_n &= a_1 + a_2 + \dots + a_n \\ &= 0 + 0 + \dots + 0 \\ &= 0 \end{aligned}$$

i.e.,  $S_n = 0$

$$\therefore \lim_{n \rightarrow \infty} S_n = \lim_{n \rightarrow \infty} 0 = 0$$

$\therefore (S_n)$  is convergent and so,  $\sum_{n=1}^{\infty} 0 = 0$

Ex 04 :- Consider  $\sum_{n=1}^{\infty} \frac{1}{n}$  Called a harmonic Series.

Let  $a_n = \frac{1}{n} \quad \forall n \in \mathbb{N}$

$$\begin{aligned} S_n &= a_1 + a_2 + a_3 + \dots + a_n \\ &= 1 + \frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{n} \quad \forall n \in \mathbb{N} \end{aligned}$$

Note that; the Sequence  $(S_n)$  is not Cauchy.

## Cauchy Sequence;

A sequence  $(a_n)$  is Cauchy if for any  $\epsilon > 0$   
 $\exists$  an  $N$  such that

$$|a_n - a_m| < \epsilon \quad \text{where } m, n > N \quad \text{Here } m, n \in \mathbb{N}$$

Claim;

Take any two positive integers  $m$  and  $n$  with  $n > m$

Now,

$$|S_n - S_m| = \left| \left( 1 + \frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{n} \right) - \left( 1 + \frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{m} \right) \right|$$

$$= \left| \frac{1}{m+1} + \frac{1}{m+2} + \dots + \frac{1}{n} \right|$$

$$= \frac{1}{m+1} + \frac{1}{m+2} + \dots + \frac{1}{n}$$

$$\geq \frac{1}{n} + \frac{1}{n} + \dots + \frac{1}{n}$$

$$= (n-m) \frac{1}{n}$$

$$= 1 - \frac{m}{n}$$

$$\text{i.e. } |S_n - S_m| \geq 1 - \frac{m}{n}$$

$\forall n, m \in \mathbb{N}$  with  $n > m$

As  $n > m$

Let  $n = 2m$

$$\text{Then } |S_n - S_m| > 1 - \frac{m}{n} = 1 - \frac{m}{2m} = \frac{1}{2} > 0$$

$\therefore (S_n)$  is not a Cauchy sequence //

$$\left. \begin{array}{l} n > m \\ \frac{1}{n} < \frac{1}{m} \\ m+1 < n \\ m+2 < n \\ m+3 < n \\ \frac{1}{m+1} > \frac{1}{n} \\ \frac{1}{m+2} > \frac{1}{n} \end{array} \right\}$$

Remark 01;

A Sequence of real numbers is Convergent, if and only if (iff) it is Cauchy,

Here,  $(S_n)$  is not Convergent.  $\therefore \sum_{n=1}^{\infty} \frac{1}{n}$  is divergent.  
Also,

$$|S_{n+1} - S_n| = (1 + \frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{n+1}) - (1 + \frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{n}) \\ = \frac{1}{n+1} > 0 \quad \forall n \in \mathbb{N}$$

$\therefore (S_n)$  is monotonic increasing and it is not bounded above.

That is  $(S_n)$  is increasing and divergent.

$$\therefore \sum_{n=1}^{\infty} \frac{1}{n} = \infty$$

Theorem 01;

If  $\sum_{n=1}^{\infty} a_n$  is Convergent then  $\lim_{n \rightarrow \infty} a_n = 0$

$$\sum_{n=1}^{\infty} a_n \xrightarrow{\text{Convergent}} \lim_{n \rightarrow \infty} a_n = 0$$

Note:- This theorem says that  $\sum_{n=1}^{\infty} a_n$  is Convergent  
 $\lim_{n \rightarrow \infty} a_n = 0$

Is the Converse of the theorem true?

If the  $\lim_{n \rightarrow \infty} a_n = 0$

Then  $\sum_{n=1}^{\infty} a_n$  is Convergent???

No!

Counter Example :-  
Consider  $\sum_{n=1}^{\infty} \frac{1}{n}$

$$\text{Let } a_n = \frac{1}{n} \quad \forall n \in \mathbb{N}$$

$$\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} \frac{1}{n} = 0$$

But  $\sum_{n=1}^{\infty} \frac{1}{n}$  is divergent.

Proof of theorem 1 ;

$$\text{Let } S_n = \underbrace{a_1 + a_2 + a_3 + \dots + a_{n-1}}_{S_{n-1}} + a_n$$

$$S_n = S_{n-1} + a_n$$

$$S_n - S_{n-1} = a_n$$

That is ;  $a_n = S_n - S_{n-1}$  ; Let  $\lim_{n \rightarrow \infty} S_n = S$

$$\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} (S_n - S_{n-1})$$

$$= \lim_{n \rightarrow \infty} S_n - \lim_{n \rightarrow \infty} S_{n-1}$$

$$= S - S$$

$$= 0$$

Since  $\sum_{n=1}^{\infty} a_n$  is convergent  $(S_n)$  is convergent.

Test for divergence ;

If  $\lim_{n \rightarrow \infty} a_n \neq 0$ , then  $\sum_{n=1}^{\infty} a_n$  is divergent.

Ex :- Consider this  $\sum_{n=1}^{\infty} n \sin(\frac{1}{n})$

$$\text{Let } a_n = n \sin(\frac{1}{n}) \quad \forall n \in \mathbb{N}$$



$$\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} n \sin\left(\frac{1}{n}\right)$$

$$= \lim_{n \rightarrow \infty} \frac{\sin\left(\frac{1}{n}\right)}{\frac{1}{n}}$$

$$\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1; \text{ where } x = \frac{1}{n}$$

$$\therefore \lim_{n \rightarrow \infty} \frac{\sin\left(\frac{1}{n}\right)}{\frac{1}{n}}$$

$$= 1 \neq 0$$

$\therefore$  The Series  $\sum_{n=1}^{\infty} n \sin\left(\frac{1}{n}\right)$  is divergent //

$$\sum_{n=1}^{\infty} \left(1 + \frac{1}{n}\right) \Rightarrow \text{divergent.}$$

Ex:- Consider this  $\sum_{n=1}^{\infty} (-1)^n$

Let  $a_n = (-1)^n \forall n \in \mathbb{N}$

$$\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} (-1)^n = \text{does not exist}$$

$$\therefore \lim_{n \rightarrow \infty} a_n \neq 0$$

$$\therefore \sum_{n=1}^{\infty} (-1)^n \text{ is divergent //}$$

\* Alternative method;

$$S_n = a_1 + a_2 + a_3 + \dots + a_n = \begin{cases} 0 & \text{if } n \text{ is even} \\ -1 & \text{if } n \text{ is odd} \end{cases}$$

$$\lim_{n \rightarrow \infty} S_n \text{ DNE } \sum_{n=1}^{\infty} (-1)^n \text{ is divergent.}$$

Ex:-

Consider the Geometric Series with Common ratio  $r$   
 $\sum_{n=1}^{\infty} r^{n-1}$

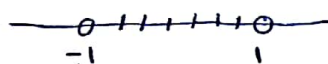
Let  $a_k = r^{k-1}$ , where  $k=1, 2, 3, \dots$

$$S_n = a_1 + a_2 + a_3 + \dots + a_n \\ = 1 + r + r^2 + \dots + r^{n-1}$$

$$= \begin{cases} \frac{1-r^n}{1-r} & \text{if } r \neq 1 \\ n & \text{if } r = 1 \end{cases}$$

Case i :  $|r| < 1 \sim -1 < r < 1 \sim r \in (-1, 1)$

$$|r| = \begin{cases} r & \text{if } r \geq 0 \\ -r & \text{if } r < 0 \end{cases}$$



$$\lim_{n \rightarrow \infty} S_n = \lim_{n \rightarrow \infty} \frac{1-r^n}{1-r} = \frac{1}{1-r}$$

$\therefore$  The geometric Series is Convergent

$$\text{i.e. } \sum_{n=1}^{\infty} r^{n-1} = \frac{1}{1-r} \quad \text{if } |r| < 1$$

Case ii :-  $|r| \geq 1$

$$\text{Let } a_n = r^{n-1}$$

$$\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} r^{n-1} \neq 0$$

$\therefore \sum_{n=1}^{\infty} a_n$  is not Convergent.

Ex:-

Consider  $x = 0.999 \dots$

$$x = 0 \times 10^0 + 9 \times 10^{-1} + 9 \times 10^{-2} + 9 \times 10^{-3} + \dots$$

$$x = \frac{9}{10} + \frac{9}{10^2} + \frac{9}{10^3} + \dots$$

$$= \frac{9}{10} \left[ 1 + \frac{1}{10} + \frac{1}{10^2} + \dots \right]$$

Geometric Series with  $r = \frac{1}{10} < 1$

$$= \frac{9}{10} \left( \frac{\frac{1}{10}}{1 - \frac{1}{10}} \right)$$

$$= \frac{9}{10} \times \frac{10}{9} \times \frac{1}{9}$$

$$= \underline{\underline{1}}$$

$$a + ar + ar^2 + \dots + ar^{n-1} = \frac{a(1-r^n)}{1-r} \quad \text{if } r \neq 1$$

$$a + ar + ar^2 + \dots = \lim_{n \rightarrow \infty} \frac{a(1-r^n)}{1-r}$$

$$= \frac{a}{1-r} \lim_{n \rightarrow \infty} (1-r^n)$$

$$= \frac{a}{1-r}$$

$$\lim_{n \rightarrow \infty} = 0$$

$$\sum_{n=1}^{\infty} ar^{n-1} = \frac{a}{1-r} \quad \text{if } r \neq 1$$

Theorem 02 :-

- a) If the Series  $\sum_{n=1}^{\infty} a_n$  and  $\sum_{n=1}^{\infty} b_n$  Converges, then  $\sum_{n=1}^{\infty} (a_n + b_n)$  Converges and

$$\sum_{n=1}^{\infty} (a_n + b_n) = \sum_{n=1}^{\infty} a_n + \sum_{n=1}^{\infty} b_n$$

- b) If the Series  $\sum_{n=1}^{\infty} a_n$  Converges and  $C$  is a Constant then the Series  $\sum_{n=1}^{\infty} (C a_n)$  Converge and

$$\sum_{n=1}^{\infty} (C a_n) = C \sum_{n=1}^{\infty} a_n$$

Proof :-

Let  $S_n$ ,  $t_n$  and  $U_n$  be the partial Sums of  $\sum_{n=1}^{\infty} a_n$ ,  $\sum_{n=1}^{\infty} b_n$  and  $\sum_{n=1}^{\infty} (a_n + b_n)$ .

$$S_n = a_1 + a_2 + a_3 + \dots + a_n$$

$$t_n = b_1 + b_2 + b_3 + \dots + b_n$$

$$U_n = c_1 + c_2 + c_3 + \dots + c_n$$

where  $c_k = a_k + b_k$ ;  $k = 1, 2, 3, \dots, n$ .

$$\begin{aligned} U_n &= (a_1 + b_1) + (a_2 + b_2) + \dots + (a_n + b_n) \\ &= (a_1 + a_2 + a_3 + \dots + a_n) + (b_1 + b_2 + b_3 + \dots + b_n) \end{aligned}$$

we know that both  $\sum_{n=1}^{\infty} a_n$  and  $\sum_{n=1}^{\infty} b_n$  are Convergent

$\therefore$  The sequences  $(S_n)$  and  $(t_n)$  are convergent,

$\therefore$  The Sequence  $(S_n + t_n)$  is also Convergent.

Sums.

$$\therefore \lim_{n \rightarrow \infty} (S_n + t_n) = \lim_{n \rightarrow \infty} S_n + \lim_{n \rightarrow \infty} t_n$$

$$= a + b \quad \text{where}$$



where ;

$$a = \lim_{n \rightarrow \alpha} S_n \text{ and } \lim_{n \rightarrow \alpha} t_n$$

$$\therefore \lim_{n \rightarrow \alpha} u_n = a + b$$

$\therefore \sum_{n=1}^{\alpha} (a_n + b_n)$  is convergent. and

$$\sum_{n=1}^{\alpha} (a_n + b_n) = \sum_{n=1}^{\alpha} a_n + \sum_{n=1}^{\alpha} b_n.$$

$$\text{Ex :- Find } \sum_{n=1}^{\infty} \frac{2^n - 4^{n+2}}{5^{n-1}}$$

$$\text{note :- } \frac{2^n - 4^{n+2}}{5^{n-1}} = 2 \left( \frac{2}{5} \right)^{n-1} - 4^3 \left( \frac{4}{5} \right)^{n-1}$$

Both the Series  $\sum_{n=1}^{\infty} \left( \frac{2}{5} \right)^{n-1}$  and  $\sum_{n=1}^{\infty} \left( \frac{4}{5} \right)^{n-1}$  are geometric Series with ratios  $r = \frac{2}{5}$  and  $r = \frac{4}{5} < 1$

Since their Common ratios are in  $(-1, 1)$  then are convergent

$$\sum_{n=1}^{\infty} \left( \frac{2}{5} \right)^{n-1} = \frac{1}{1 - \frac{2}{5}} = \frac{5}{3} \text{ and } \sum_{n=1}^{\infty} \left( \frac{4}{5} \right)^{n-1} = \frac{1}{1 - \frac{4}{5}} = 5$$

$$\therefore \sum_{n=1}^{\infty} \frac{2^n - 4^{n+2}}{5^{n-1}}$$

$$= 2 \sum_{n=1}^{\infty} \left( \frac{2}{5} \right)^{n-1} - 4^3 \sum_{n=1}^{\infty} \left( \frac{4}{5} \right)^{n-1}$$

$$= 2 \times \frac{5}{3} - 4^3 \times 5$$

$$= \frac{10}{3} - 320$$

$$= -\frac{950}{3}$$

$(x_n)$  is bounded if  $\exists M$  s.t.  $|x_n| < M$   
 $M > 0$  such that  $|x_n| < M$

Theorem 03 :-

Let  $\sum_{n=1}^{\infty} a_n$  be an infinite series of nonnegative terms.  $a_n \geq 0 \forall n \in \mathbb{N}$ .

Then  $\sum_{n=1}^{\infty} a_n$  converges if and only if the sequence of partial sums is bounded

i.e. Let  $S_n = \sum_{k=1}^n a_k$ .  $\sum_{n=1}^{\infty} a_n$  is convergent iff  $(S_n)$  is bounded.

Proof :-

Let  $S_n = a_1 + a_2 + a_3 + \dots + a_n = \sum_{k=1}^n a_k$  ①  $\forall n \in \mathbb{N}$

①  $\Rightarrow S_{n+1} = a_1 + a_2 + a_3 + \dots + a_n + a_{n+1}$

$$S_{n+1} = S_n + a_{n+1}$$

$$S_{n+1} - S_n = a_{n+1} \geq 0 \quad \forall n \in \mathbb{N}$$

$$\therefore S_{n+1} - S_n \geq 0$$

That is,  $S_{n+1} \geq S_n \quad \forall n \in \mathbb{N}$

$$S_1 \leq S_2 \leq S_3 \dots$$

So, the sequence  $(S_n)$  is monotonically increasing.  
By the Monotonic Convergence theorem (MCT)

The sequence  $(S_n)$  is convergent iff

$(S_n)$  is bounded.

i.e.  $\sum_{n=1}^{\infty} a_n$  is convergent iff  $(S_n)$  is bounded  
(bdd)

Theorem 04 :-

The infinite Series  $\sum_{n=1}^{\infty} a_n$  Converges iff for each  $\epsilon > 0$   $\exists$  a positive integer  $N(\epsilon)$  Such that ;

$$|a_{n+1} + a_{n+2} + \dots + a_m| < \epsilon \quad \text{whenever } m > n > N(\epsilon)$$

Theorem 05 :- (Comparison's test)

Let  $\sum_{n=1}^{\infty} a_n$  and  $\sum_{n=1}^{\infty} b_n$  be infinite Series of non-negative terms  $a_n$  and  $b_n$  with  $a_n \leq b_n$

a) If  $\sum_{n=1}^{\infty} b_n$  Converges, then  $\sum_{n=1}^{\infty} a_n$  Converges.

b) If  $\sum_{n=1}^{\infty} a_n$  diverges, then  $\sum_{n=1}^{\infty} b_n$  diverges.

$$a_n, b_n \geq 0 \quad \text{and} \quad a_n \leq b_n$$

$$\overline{\quad \quad \quad} \quad \overline{\quad \quad \quad}$$

$a_n \qquad \qquad b_n$

$$\begin{array}{ccc} \sum a_n & & \sum b_n \\ \text{cgt} & \longleftarrow & \text{cgt} \\ \text{dug} & \longrightarrow & \text{dug} \end{array}$$

Ex:- Show that  $\sum_{n=1}^{\infty} \frac{1}{n^2}$  is Convergent.

Let  $a_n = \frac{1}{n^2} \quad \forall n \in \mathbb{N}$

$$a_n = \frac{1}{n^2} = \left(\frac{1}{n}\right)\left(\frac{1}{n}\right) < \frac{1}{n} \frac{1}{(n-1)}$$

$$= \frac{1}{n(n-1)} \quad \forall n = 2, 3, 4, \dots$$

$$\text{Let } b_n = \frac{1}{n(n-1)} \quad \forall n \geq 2.$$

$$b_n = \frac{-1}{n} + \frac{1}{n-1}$$

$$b_2 = \frac{-1}{2} + \frac{1}{1}$$

$$b_3 = \frac{-1}{3} + \frac{1}{2}$$

$$\vdots$$

$$b_{n-1} = \frac{-1}{n-1} + \frac{1}{n}$$

$$b_n = \frac{-1}{n} + \frac{1}{n-1}$$

$$\therefore \sum_{k=2}^n b_k = 1 - \frac{1}{n}$$

$$\lim_{n \rightarrow \infty} \sum_{k=2}^n b_k = \lim_{k \rightarrow \infty} \left(1 - \frac{1}{k}\right) = 1$$

$$\Rightarrow \sum_{n=2}^{\infty} b_n = 1$$

$$\therefore \sum_{n=1}^{\infty} b_n = b_1 + \sum_{n=2}^{\infty} b_n$$

$$= b_1 + 1$$

$$\therefore \sum_{n=1}^{\infty} b_n \text{ is Convergent.}$$

$\therefore$  By Comparison's test,

$$\sum_{n=1}^{\infty} a_n = \sum_{n=1}^{\infty} \frac{1}{n^2} \text{ is Convergent.}$$



# CT = Comparison's Test.

EX:- Show that  $\sum_{n=1}^{\infty} \frac{2^n}{3^n(n+1)^2+9}$  is Convergent.

$$\text{Let } a_n = \frac{2^n}{3^n(n+1)^2+9} > 0 \quad \forall n \in \mathbb{N}$$

$$a_n = \frac{2^n}{3^n(n+1)^2+9} > b_n$$

$$a_n = \frac{2^n}{3^n(n+1)^2+9} < \frac{2^n}{3^n} = \left(\frac{2}{3}\right)^n \quad \forall n \in \mathbb{N}$$

$$\text{Let } b_n = \left(\frac{2}{3}\right)^n$$

$$\begin{aligned} \sum_{n=1}^{\infty} b_n &= \sum_{n=1}^{\infty} \left(\frac{2}{3}\right)^n = \sum_{n=1}^{\infty} \left(\frac{2}{3}\right) \left(\frac{2}{3}\right)^{n-1} \\ &= \frac{2}{3} \sum_{n=1}^{\infty} \left(\frac{2}{3}\right)^{n-1} \end{aligned}$$

A geometric Series

with Common ratio  $r = \frac{2}{3} < 1$

$\therefore$  It is Convergent,

Now by the CT,  $\sum_{n=1}^{\infty} a_n$  is Convergent,

ex:- Consider this  $\sum_{n=1}^{\infty} \frac{1}{5n-3}$

$$\text{Let } c_n = \frac{1}{5n-3} > 0 \quad \forall n \in \mathbb{N}$$

$$\text{Note that } \frac{1}{5n-3} > \frac{1}{5n} \quad \forall n \in \mathbb{N}$$

$$\text{Let } a_n = \frac{1}{5n} \text{ and } b_n = \frac{1}{5n-3} \quad \forall n \in \mathbb{N} \quad \sum \frac{1}{5n}$$

Since  $\sum_{n=1}^{\infty} \frac{1}{n}$  is divergent,  $\sum_{n=1}^{\infty} a_n$  is divergent.

$\therefore \sum_{n=1}^{\infty} b_n$  is divergent. //

ex:- Consider  $\sum_{n=1}^{\infty} \frac{1}{2n-15} <$

$$2n-15 > 0$$

$$n > 7.5$$

$$a_n > 0 \text{ if } n \geq 8.$$

$$\therefore \sum_{n=1}^{\infty} a_n = (a_1 + a_2 + a_3 + a_4 + a_5 + a_6 + a_7) + \sum_{n=8}^{\infty} \frac{1}{2n-15}$$

$$\text{Let } a_n = \frac{1}{2n-15}$$

$$a_1 = -\frac{1}{13} < 0$$

$$a_5 = -\frac{1}{5} < 0$$

$$a_2 = -\frac{1}{11} < 0$$

$$a_6 = -\frac{1}{9} < 0$$

$$a_3 = -\frac{1}{9} < 0$$

$$a_7 = -\frac{1}{1} < 0$$

$$a_4 = -\frac{1}{7} < 0$$

$$a_8 = \frac{1}{1} > 0$$

Note that  $\frac{1}{2n-15} > 0$  ~~for~~  $n \geq 8$ .

$$\frac{1}{2n-15} > \frac{1}{2n} \text{ for } n \geq 8.$$

Then  $a_n > a'_n$

$$\text{Let } a'_n = \frac{1}{2n}$$

Since  $\sum_{n=1}^{\infty} a'_n$  is divergent, by CT,

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$\sum_{n=8}^{\infty} a_n$  is divergent.

$\therefore \sum_{n=1}^{\infty} a_n$  is divergent.

**Definition :-**

If  $\sum_{n=1}^{\infty} |a_n|$  Converges, then the series  $\sum_{n=1}^{\infty} a_n$  is said to be absolutely Convergent.

Ex:-

Consider  $\sum_{n=1}^{\infty} \frac{(-1)^n}{n^2}$

Let  $a_n = \frac{(-1)^n}{n^2} \quad \forall n \in \mathbb{N}$

$$\Rightarrow |a_n| = \left| \frac{(-1)^n}{n^2} \right| = \frac{1}{n^2} |(-1)^n| = \frac{1}{n^2}$$

We know that the series  $\sum_{n=1}^{\infty} \frac{1}{n^2}$  is convergent.

$\therefore$  The series  $\sum_{n=1}^{\infty} \frac{(-1)^n}{n^2}$  is absolutely Convergent.

**Theorem 6 :-**

If a series is absolutely Convergent.  
Then it is Convergent.

$$\sum_{n=1}^{\infty} |a_n| \xrightarrow{\text{cgt}} \sum_{n=1}^{\infty} a_n \text{ cgt.}$$

$$\sum_{n=1}^{\infty} a_n \not\xrightarrow{\text{cgt}} \sum_{n=1}^{\infty} |a_n| \text{ cgt.}$$

Ex :-

Consider  $\sum_{n=1}^{\infty} \frac{(-1)^n}{n}$

We shall later show that the series  $\sum_{n=1}^{\infty} \frac{(-1)^n}{n}$  is convergent.

Let  $a_n = \frac{(-1)^n}{n}$

Then  $|a_n| = \left| \frac{(-1)^n}{n} \right| = \frac{1}{n}$

Also we know that Series  $\sum_{n=1}^{\infty} \frac{1}{n}$  is divergent.  
i.e; the Series  $\sum_{n=1}^{\infty} \frac{(-1)^n}{n}$  is convergent, but  
Not it is absolutely convergent.

**Theorem 07 :- (Limit form of CT)**

Let  $\sum_{n=1}^{\infty} a_n$  and  $\sum_{n=1}^{\infty} b_n$  be infinite Series

of Positive terms; (i.e,  $a_n, b_n > 0$ )

Such that; ~~limit~~

$$\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = l.$$

if  $0 < l < \infty$ , then two Series Converge or  
diverge together

$$\sum a_n \quad \sum b_n$$

$$cgt \Rightarrow cgt$$

$$cgt \Leftarrow cgt$$

$$dvg \Rightarrow dvg$$

$$dvg \Leftarrow dvg$$



Ex:-

Consider  $\sum_{n=1}^{\infty} \frac{n}{8n^3+9}$

Let  $a_n = \frac{n}{8n^3+9} > 0 \quad \forall n \in \mathbb{N}$

Note:-

$$\frac{1}{n^2} > \frac{n}{8n^3} > \frac{n}{8n^3+9}$$

Let  $b_n = \frac{1}{n^2} > 0 \quad \forall n \in \mathbb{N}$

$$L = \lim_{n \rightarrow \infty} \frac{a_n}{b_n} = \lim_{n \rightarrow \infty} \left[ \frac{\frac{n}{8n^3+9}}{\frac{1}{n^2}} \right]$$

$$= \lim_{n \rightarrow \infty} \frac{n^3}{8n^3+9}$$

$$= \lim_{n \rightarrow \infty} \frac{1}{8 + \frac{9}{n^3}}$$

$$= \frac{1}{8} > 0$$

Also, the Series  $\sum_{n=1}^{\infty} \frac{1}{n^2}$  is Convergent.

$\therefore$  by CT, the Series  $\sum_{n=1}^{\infty} \frac{n}{8n^3+9}$  is Convergent.

Ex:- Consider  $\sum_{n=1}^{\infty} \frac{n}{8n^3-9}$

$a_n > 0$  iff  $8n^3-9 > 0$

$$8n^3 > 9$$

$$n > \frac{9^{1/3}}{2}$$

$\therefore a_n > 0 \quad \forall n \geq 2$

Let  $a_n = \frac{n}{8n^3 - 9}$

$a_1 = \frac{1}{8-9} = -1$

$\sum_{n=1}^{\infty} a_n = a_1 + \sum_{n=2}^{\infty} a_n$

$\underbrace{\left(\frac{n}{8n^3-9}\right)}_{b_n} > \frac{n}{8n^3} = \underbrace{\left(\frac{1}{8n^2}\right)}_{a_n} = a'_n \text{ (Say)}$

$$L = \lim_{n \rightarrow \infty} \frac{a_n}{b_n} = \lim_{n \rightarrow \infty} \frac{\frac{1}{8n^2}}{\frac{n}{8n^3-9}}$$

$$= \lim_{n \rightarrow \infty} \frac{8n^3-9}{8n^3}$$

$$= \lim_{n \rightarrow \infty} \frac{8 - \frac{9}{n^3}}{8}$$

$$= 1 > 0 \quad (*)$$

But we know that  $\sum_{n=1}^{\infty} \frac{1}{n^2}$  is convergent  $(*)$ .

$\therefore$  By CT, the given Series is Convergent.

Determine the following Series are Convergent or divergent.

i)  $\sum_{n=1}^{\infty} \sin\left(\frac{1}{n}\right)$

(ii)  $\sum_{n=1}^{\infty} \frac{\sin(n\theta)}{n^2}$ , where  $\theta \in \mathbb{R}$ .

### Theorem 08 :- (Integral test).

Let  $f$  be a non negative decreasing Continuous function defined on  $[1, \infty)$ . Then the  $\sum_{n=1}^{\infty} f_n$  Converges iff.

$\lim_{n \rightarrow \infty} \int_1^n f(x) dx$  exists as a real number

(i)  $f(x) \geq 0 \quad \forall x \in [1, \infty)$

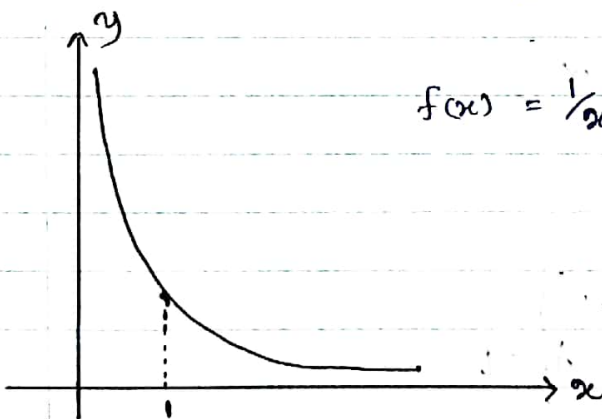
(ii)  $f$  is  $\searrow$  on  $[1, \infty)$

(iii)  $f$  is Continuous on  $[1, \infty)$   $f(x) = \frac{1}{x^2 + 2x}$

$\sum_{n=1}^{\infty} \frac{1}{n^2 + 2n}$   $f_n = f(n)$   
 $f_n = \frac{1}{n^2 + 2n}$

Ex:- Use the integral test to show that  $\sum_{n=1}^{\infty} \frac{1}{n}$  is divergent.

So, Let  $f(x) = \frac{1}{x}$  for  $x > 0$



(i)  $f(x) > 0 \quad \forall x \in [1, \infty)$

(ii)  $f(x)$  is decreasing on  $[1, \infty)$

(iii)  $f(x)$  is Continuous on  $[1, \infty)$

∴ we may apply the integral test

$$\lim_{n \rightarrow \infty} \int_1^n f(x) dx = \lim_{n \rightarrow \infty} \left[ \int_1^n \frac{1}{x} dx \right]$$

$$= \lim_{n \rightarrow \infty} \left[ \ln(x) \right]_{x=1}^n$$

$$= \lim_{n \rightarrow \infty} \left[ \ln n - \ln 1 \right]$$

$$= \infty$$

Thus, by the integral test the given Series is divergent.

Ex:- Consider  $\sum_{n=2}^{\infty} \frac{1}{n \ln(n)}$

Let  $f(x) = \frac{1}{x \ln(x)}$ ,  $x > 1$

Then

(i)  $f(x) > 0 \quad \forall x \in [2, \infty)$

(ii)  $f(x) \searrow$  on  $[2, \infty)$

(iii)  $f$  is cts on  $[2, \infty)$

Now by the integral test.

$$\begin{aligned} \lim_{n \rightarrow \infty} \int_2^n \frac{1}{x \ln(x)} dx &= \lim_{n \rightarrow \infty} \int_2^n \frac{1}{\ln(x)} d[\ln(x)] \\ &= \lim_{n \rightarrow \infty} \left[ \ln(\ln(x)) \right]_{x=2}^n \end{aligned}$$



$$= \lim_{n \rightarrow \infty} \left[ \ln(\ln(n)) - \ln(\ln(2)) \right]$$

$$= \infty$$

$\therefore$  By the integral the given Series is divergent.

(1) HW: Consider the Series  $\sum_{n=1}^{\infty} \frac{1}{n^p}$  where,  $p \in \mathbb{R}$

This is called P-Series, use the Integral test to determine the diverges or Convergent

- i) Case 1 ;  $p < 0$  div
- ii) Case 2 ;  $p = 0$  div
- iii) Case 3 ;  $0 < p < 1$  div
- iv) Case 4 ;  $p = 1$  div
- v) Case 5 ;  $p > 1$  cgt.

(2) Use P-Series test to determine the Converges of the following

(i)  $\sum_{n=1}^{\infty} \frac{1}{\sqrt{n}}$  ~~div~~

(ii)  $\sum_{n=1}^{\infty} \frac{1}{n^{3/2}}$

(3) Show that the Series  $\sum_{n=1}^{\infty} \frac{(-1)^n (n^2 + 3)}{n^6 + 5n^3 + 7}$  is Convergent  
Limit form of.  
(use P-Series + nCT)

(4) Use the Integral test to determine the Converges/divergence of the Series,  $\sum_{n=2}^{\infty} \frac{1}{n \ln(n) \ln(\ln(n))}$

### Theorem 09 (Ratio test).

Let  $\sum a_n$  be a Series with non-zero terms.

- (a) If there exist a positive number  $r < 1$  and a positive integer,  $N \in \mathbb{N}$  such that

$$\left| \frac{a_{n+1}}{a_n} \right| \leq r < 1 \quad \forall n \geq N$$

Then  $\sum a_n$  is absolutely Convergent.  
(i.e., it is Convergent)

- (b) if there exist a number  $r > 1$  and a positive integer  $N \in \mathbb{N}$  such that

$$\left| \frac{a_{n+1}}{a_n} \right| \geq r > 1 \quad n > N, \text{ then } \sum a_n \text{ is divergent.}$$

### Theorem 10. (Limit form of ratio test).

Let  $\sum_{n=1}^{\infty} a_n$  be a Series with non-zero terms

$$\text{Let } \lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = L.$$

- (a) if  $L < 1$ , then  $\sum_{n=1}^{\infty} a_n$  is absolutely Convergent  
(that is Convergent)

- (b) if  $L > 1$ , then  $\sum_{n=1}^{\infty} a_n$  is divergent.

(c) if  $\ell = 1$ , then the test is inconclusive,